

GROUP ASSESSMENT / GROUP 7

QUESTION 1. Relation Schema for the MemberDinner Table

MemberDinner (MEMBER NUM, MEMBER NAME, MEMBER ADDRESS, DINNER NUM, DINNER DATE, VENUE CODE, VENUE DESCRIPTION, FOOD CODE, FOOD DESCRIPTION)

QUESTION 2. For a table to be in 1NF, it must contain only atomic values. We eliminate the repeating groups by splitting rows with multiple food items into separate, distinct rows. And each column contains values with same data type.

MEMBER NUM	MEMBER NAME	MEMBER ADDRESS	DINNER NUM	DINNER DATE	VENUE CODE	VENUE DESCRIPTION	FOOD CODE	FOOD DESCRIPTION
214	Bella Uwimana	Kicukiro	D0001	15-May-20	B01	Grand Ball Room	EN3,	Stuffed crab,
214	Bella Uwimana	Kicukiro	D0001	15-May-20	B01	Grand Ball Room	DE8	Chocolate mousse
331	Bosco Rukundo	Masaka	D0002	11-Nov-20	B02	Petit Ball Room	EN5	Marinated steak
331	Bosco Rukundo	Masaka	D0002	11-Nov-20	B02	Petit Ball Room	DE8	Chocolate mousse
189	Florence Kanyana	Nyaruge nge	D0003	11-Nov-20	C01	Café	SO1	Pumpkin soup
189	Florence Kanyana	Nyaruge nge	D0003	11-Nov-20	C01	Café	EN5	Marinated steak
189	Florence Kanyana	Nyaruge nge	D0003	11-Nov-20	C01	Café	DE2	Apple pie
124	Kevin Ganza	Kanombe	D0003	18-Nov-20	C01	Café	SO1	Pumpkin soup,
124	Kevin Ganza	Kanombe	D0003	18-Nov-20	C01	Café	ENE 5	Marinated steak,
124	Kevin Ganza	Kanombe	D0003	18-Nov-20	C01	Café	DE2	Apple pie
310	Emma Kaliza	Gikondo	D0004	18-Nov-20	E10	Petit Ball Room	DE2	Apple Pie

QUESTION 3. To reach 2NF, the table must be in 1NF and have no partial dependencies (no non-prime attribute should depend on only a part of the composite primary key).

- MEMBER NAME, MEMBER ADDRESS, DINNER NUM, DINNER DATE, VENUE CODE, and VENUE DESCRIPTION depend solely on MEMBER_NUM (Partial Dependency).
- FOOD DESCRIPTION depends solely on FOOD CODE (Partial Dependency).
- The link between a member and what they ate depends on both elements: (MEMBER NUM, FOOD CODE).

Table A: Member Dinner Details

Primary Key: MEMBER NUM

MEMBER NUM	MEMBER NAME	MEMBER ADDRESS	DINNER NUM	DINNER DATE	VENUE CODE	VENUE DESCRIPTION
214	Bella Uwimana	Kicukiro	D0001	15-May-20	B01	Grand Ball Room
331	Bosco Rukundo	Masaka	D0002	11-Nov-20	B02	Petit Ball Room
189	Florence Kanyana	Nyarugenge	D0003	11-Nov-20	C01	Café
124	Kevin Ganza	Kanombe	D0003	18-Nov-20	C01	Café
310	Emma Kaliza	Gikondo	D0004	18-Nov-20	E10	Petit Ball Room

Table B: Food

Primary Key: FOOD CODE

FOOD CODE	FOOD DESCRIPTION
EN3	Stuffed crab
DE8	Chocolate mousse
EN5	Marinated steak
SO1	Pumpkin soup
DE2	Apple pie
ENE5	Marinated steak

Table C: Member Food

Primary Key: (MEMBER NUM, FOOD CODE)

MEMBER NUM	FOOD CODE
214	EN3
214	DE8
331	EN5
331	DE8
189	SO1
189	EN5
189	DE2
124	SO1
124	ENE5
124	DE2
310	DE2

QUESTION 4. 3rd Normal Form (3NF) Tables

This table stores information about members only. Nothing else just who the members are and where they live.

Table 1: Member

Member Num(PK)	Member Name	Member Address
124	Kevin Ganza	Kanombe
189	Florence Kanyama	Nyarugenge
214	Bella Uwimana	Kicukiro
310	Emma Kaliza	Gikondo
331	Bosco Rukundo	Masaka

Schema: Member(Member Num, Member Name, Member Address)

This table stores information about venues the places where dinners are held

Table 2: Venue

Venue Code (PK)	Venue Description
B01	Grand Ball Room
B02	Petit Ball Room
C01	Café
E10	Petit Ball Room

Schema: Venue(Venue Code, Venue Description)

This table stores information about dinner events when each dinner happened and where it was held.

Table 3: Dinner

Dinner Num (PK)	Dinner Date	Venue Code (FK → Venue)
D0001	15-Nov-20	B01
D0002	11-Nov-20	B02
D0003	11-Nov-20	C01
D0004	18-Nov-20	E10

Schema: Dinner(Dinner Num, Dinner Date, Venue Code)

This table stores information about food items available at dinners.

Table 4: Food

Food Code (PK)	Food Description
DE2	Apple pie
DE8	Chocolate mousse
EN3	Stuffed crab
EN5	Marinated steak

SO1	Pumpkin soup
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Schema: Food(Food Code, Food Description)

This table is the connector — it links members, dinners, and food together. It answers the question:

Which member attended which dinner and chose which food?

Table 5: Member Dinner Food (Junction Table)

Member Num (PK, FK → Member)	Dinner Num (PK, FK → Dinner)	Food Code (PK, FK → Food)
214	D0001	EN3
214	D0001	DE8
331	D0002	EN5
331	D0002	DE8
189	D0003	SO1
189	D0003	EN5
189	D0003	DE2
124	D0003	SO1
124	D0003	EN5
124	D0003	DE2
310	D0004	DE2

Schema: Member Dinner Food(Member Num, Dinner Num, Food Code)

- **Member, Venue, Food** have a simple single-column primary key and one or two non-key attributes that depend directly and only on that key with no transitive chains possible
- **Dinner** had its transitive dependency (Venue Code → Venue Description) removed by delegating Venue Description to the Venue table, leaving only direct dependencies on Dinner Num
- **Member Dinner Food** contains no non-key attributes at all, making it automatically free from both partial and transitive dependencies the purest form of a 3NF table.

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